INCH-POUND
MIL-I-46625C
31 January 1990
SUPERSEDING
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12 June 1969 and
AMENDMENT 2
20 July 1973

MILITARY SPECIFICATION

INITIATOR, CARTRIDGE ACTUATED, M3A2, ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M3A2 Cartridge Actuated Initiator (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-2550	Ammunition, General Specification For
MIL-I-14802	Initiator, Cartridge Actuated, M3A2, Remanufactured Metal Parts Assembly
MIL-I-46559	Initiator, Cartridge Actuated, M3A2, Metal Parts Assembly

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-453	Inspection, Radiographic
MS28741	Hose Assembly, Detachable End Fitting, Medium Pressure

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from Military Specifications and Standards, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 1377

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2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the soliciation.

DRAWINGS

NAVAL ORDNANCE STATION, INDIAN HEAD, MD (CAGE Code 19200 and 19203)

DL8595821	Initiator, Cartridge Actuated, M3A2 Assembly
8594157	Cartridge, Impulse, M73 Assembly
8796541	Box, Fiberboard (Inner) Packing, Ammunition, for Initiators, Cartridge Actuated, M3A2 or M5A2
8796542	Box, Packing, Ammunition, for Initiator, Cartridge Actuated, M3A2 or M5A2
9216934	Box, Fiberboard (Outer) Alternative Packing, Ammunition, for Initiator, Cartridge Actuated, M3A2 or M5A2

U.S. AIR FORCE

TPO 00-932-5031	Initiator, Cartridge Actuated, M3A2

(Application for copies should be addressed to: Commanding Officer, Naval Ordnance Station, Data Control Branch (3720), Indian Head, MD 20640-5000.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. All materials used in the manufacture of the initiators shall conform strictly to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2 Primary components. For the purpose of this specification, the M73 impulse cartridge (Drawing 8594157) is defined as a primary component. Only primary components from a single lot shall be used in a production lot of initiators. One primary component production lot may be used in more than one initiator production lot.

3.3 Firing pin release. The firing pin of the assembled initiator, less safety pin, cartridge, cartridge retainer, and chamber, shall release from the cocked position when the application of an axial load of 20 to 35 pounds has withdrawn the initiator pin 3/4 inch, maximum (max).

3.4 Initiator pin security. The initiator pin of the assembled initiator, less cartridge, cartridge retainer, and chamber, shall not withdraw beyond the safety pin when subjected to an axial load of 200 to 220 pounds. Application of the load shall not cause permanent deformation of the safety pin.

3.5 Breakaway torque. The initiator chamber and cap of the assembled initiator shall withstand a breakaway torque application of 400 inch-pounds without loosening or deformation (see 4.2.3.1).

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3.6 Residual magnetism. The initiator shall not deflect the indicator of the compass more than five degrees in either direction using the procedures in 4.3.2.

3.7 Functional.

3.7.1 Ballistics. Initiators loaded with accepted cartridges and temperature conditioned shall perform within the pressure values and ignition delays specified in table I. Pressure values shall be recorded at the end of 15 feet of aircraft hose, assembly number MS28741-4.

TABLE I. *Pressure values.*

Temperature (°F)	Pressure (psi)		Ignition delay, max, milliseconds (see 6.5.2)
	Minimum	Maximum	
-60 to -70	550	1450	65
65 to 75	700	1550	65
195 to 205	700	1700	65

3.7.2 Misfires. There shall be no misfires.

3.7.3 Mechanical failures. The initiator shall function without mechanical failure (see 6.5.1).

3.8 Workmanship. Initiators shall be free of cracks, splits, incomplete or damaged threads on inlet and outlet ends, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.8.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.8.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.8.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

3.9 Radiographic examination. Each initiator assembly shall be radiographically examined for defects as specified in 4.4.

3.10 Component parts. All component parts shall meet the requirements of MIL-I-14802 or MIL-I-46559 prior to assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.3 Inspection. Inspection shall be in accordance with MIL-A-2550 and MIL-STD-453.

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 68 to 77 °F
b. Altitude: Sea level
c. Humidity: 50 percent relative, max
d. Atmospheric pressure: 14.7 psia

4.2.1 Lot.

4.2.1.1 Submission of product. Unless otherwise specified by the contracting activity, formation and identification of lots shall be in accordance with MIL-STD-105.

4.2.1.2 Size of lot. A lot shall consist of an identifiable group of assembled initiators containing metal parts assemblies produced by a homogeneous process and loaded with one lot of accepted cartridges.

4.2.2 Examination. One-hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed in accordance with the classification of defects. Four power (4X) magnification may be used for inspection of such defects as burr and foreign matter.

4.2.2.1 Cartridge, Impulse M73, Drawing 8594157, covering a detail of Drawing 8595821 or 10524101.

Characteristic	Defect	Method of Inspection
Critical		
1	Mixed cartridges	Visual
2	Metal defective	Visual
3	Marking missing, incorrect, or illegible	Visual
Major		
101	Foreign matter present	Visual
Minor		
201	Case scratched or dented	Visual

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4.2.2.2 Assembly, Drawing 10524101 or 8595821 less Drawing 8595404, 8593013, 8593014, and 8359536.

Characteristic	Defect	Method of Inspection
Critical		
1	Mixed initiators	Visual
2	Initiator not in cocked position	Visual
3	Safety pin missing or improperly engaged	Visual
4	O-ring missing	Visual
Major		
101	Thread damaged	Visual
102	Foreign matter present	Visual
Minor		
201	O-ring not lubricated	Visual

4.2.2.3 Assembly, Drawing 10524101 or 8595821, less Drawings 8593014 and MIL-C-5501/11.

Critical		
1	Cartridge loose	Visual/Manual
2	Cartridge retainer loose	Visual/Manual
3	Metering groove of retainer missing	Visual
4	Elongated hole of cartridge retainer missing	Visual
Major		
None defined		
Minor		
None defined		

4.2.2.4 Assembly, Drawing 10524101 or Drawing 8595821.

Critical		
1	Location of cartridge retainer	Gage/x-ray
2	Any part missing	Visual/x-ray
3	Improper assembly	Visual/x-ray
4	Foreign matter in gas port	Visual
Major		
101	Length from cap boss face to gas port face, max	Gage
102	Height of cap, max	Gage
103	Cap set screw missing, loose or above flush	Visual
Minor		
201	Total length, max	Gage
202	Location of marking incorrect	Visual
203	Protective finish inadequate or defective	Visual
204	Protective cap missing	Visual

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4.2.2.5 Box, Fiberboard (Inner) Packing Ammunition, Drawing 8796541.

Characteristic	Defect	Method of Inspection
Critical		
None defined		
Major		
None defined		
Minor		
201	Gap between body and cover, max	Manual
202	Contents loose	Manual
203	Metal ends loose or distorted	Manual
204	Cut or damaged through all impregnated layers	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Tape incomplete, torn, or badly wrinkled	Visual
207	Length of tear tab, min	Visual
208	Color of tape incorrect	Visual

4.2.2.6 Box, Packing Ammunition, Drawing 8796542.

Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Manual
202	Strapping missing, loose, broken, or improperly assembled	Manual/Visual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Number of units incorrect	Visual

4.2.2.7 Box, Fiberboard (Outer) Alternative Packing Ammunition, Drawing 9216934.

Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Manual
202	Box broken or damaged	Visual
203	Number of units contained incorrect	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Tape incomplete	Visual

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4.2.3 Inspection testing. The following tests shall be performed in accordance with 4.3.

4.2.3.1 Breakaway torque. Each assembly shall be subjected to the breakaway torque test of the threaded joint between the cap and the chamber. Failure of any assembly to comply with 3.5 shall be cause for rejection of the assembly.

4.2.3.2 Residual magnetism. Each initiator shall be subjected to the test for excessive residual magnetism as specified in 4.3.2. Failure of any initiator to comply with 3.6 shall be cause for rejection of the initiator.

4.2.3.3 Firing pin release. The initiator pin of each subassembly shall be subjected to the firing-pin release test. Failure of any subassembly to comply with 3.3 shall be cause for rejection of the subassembly.

4.2.3.4 Initiator pin security. The initiator pin of each subassembly shall be subjected to the initiator pin security test. Failure of any unit to comply with 3.4 shall be cause for rejection of the subassembly.

4.2.3.5 Functional tests. The following tests shall be performed by the Government.

4.2.3.5.1 Ballistics. Twenty-one units shall be tested to determine compliance with 3.7.1. The units shall be temperature conditioned as provided in table II and 4.3.5.2. Failure of any unit to comply with 3.7.1 shall be cause for rejection of the lot.

TABLE II. *Conditioning temperature.*

Temperature (°F)	Sample size
-60 to -70	7
65 to 75	7
195 to 205	7

4.2.3.5.2 Retest. There shall be no retests.

4.2.3.5.3 Test failure. If a test failure is attributable to an assignable cause, excluding the test initiator, the original test results shall be discarded and that test reconducted.

4.3 Test methods and procedures.

4.3.1 Breakaway torque. A torque wrench set at the torque requirement specified in 3.5 shall be used in conjunction with a fixture to determine the adequacy of the joints. The chamber shall be secured in an air vise and the torque shall be applied to the cap.

4.3.2 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed not greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis, and moved slowly back past the compass. This procedure shall be repeated for each 90 degrees rotation of the assembly for four full passes.

4.3.3 Firing pin release. The subassembly shall be secured in the fixture. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the fixture which registers the amount of pull.

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4.3.4 Initiator pin security. The subassembly shall be secured in the fixture with the safety pin in place. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with a fixture which registers the amount of pull. The test shall be performed with the safety pin in both positions.

4.3.5 Functional.

4.3.5.1 Ballistics. The test sample shall be rigidly supported in the fixture. The sample shall be tested within 5 minutes after removal from the conditioning chamber. When this time limit is exceeded, the sample shall be reconditioned in accordance with 4.3.5.2. Required pressure-time readings shall be taken as specified in 3.7.1.

4.3.5.2 Temperature conditioning. Each assembled initiator shall be conditioned for a period of time not less than 4 hours nor greater than 24 hours. When reconditioning initiators at −60 °F, all condensation shall be removed from the interior surfaces before returning the initiator to the conditioning chamber.

4.4 Radiographic examination. All assembled initiators shall be radiographically examined in accordance with MIL-STD-453, with the exception that an assembled M3A2 initiator will be used for the penetrameter. Any observable imperfections in components or assembly shall be cause for rejection. The initiators shall be positioned for the most revealing exposure, with the long axis of each assembly perpendicular to the rays of the X-ray machine. All initators shall be identified with serial numbers prior to X-ray examination. The serial numbers shall be in consecutive numerical order beginning with the number 001 in each production lot. Any missing serial numbers are to be identified on the X-rays. The initiators shall be arranged on boards or trays in consecutive order and each radiograph shall carry a permanent identification of the units displayed thereon. The radiographic plate identification shall include item nomenclature, the complete lot number assigned to that lot, the span of serial numbers displayed, and the contract number. Radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test. Defective initiators found during X-ray review are to be marked on the X-ray prior to shipment and those defective items shall be removed from the production lot. Film interpretation of the radiographs shall verify the presence and correct positioning of the following parts:

a. O-ring	P/N MS28755-214
b. Steel balls present (3 ea)	P/N MS19060-3
c. Firing pin	P/N 8596168
d. Spring initiator	P/N 8595406
e. Seal	P/N 10524102
f. Firing pin housing	P/N 8596169
g. Retainer, cartridge	P/N 8596013
h. Cap set screws, No. 10 (.190)	
i. Cartridge, Impulse, M73	P/N 8594157
j. Pin, initiator	P/N 8595403
k. Chamber	P/N 8593014
l. Cap	P/N 8595404

4.5 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with section 5.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be levels A or B as specified by the contracting activity:

Level A - In accordance with Drawings 8796541 and 8796542
Level B - In accordance with Drawings 8796541 and 9216934

5.2 Air Force packaging. Air Force packaging shall be in accordance with TPO 00-932-5031.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The M3A2 initiator assembly, loaded with the M73 impulse cartridge, is used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents reference (see 2.1)
c. The responsible inspection activity (see 4.1)
d. Production lot size and test activity (see 4.2.1)
e. Direction for shipping radiographs of entire production lot along with ballistic sample to the activity conducting production lot acceptance tests (see 4.4)
f. Level of marking and packing (see 5.1)
g. Safety precautions (see 6.6).

6.3 Explosive hazard classification information.

Hazard Classification/Division: 1.4
Storage Compatibility Group: S
DOT Class: C
DOT Marking — Explosive Power Device — Class C — Handle Carefully — Keep Fire Away

6.4 Hazard notice. The initiator described herein and certain of its components are flammable, explosive, or both. Consequently it presents hazards in manufacture, handling, storage, and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling, and electrically induced incidents.

6.5 Definitions.

6.5.1 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.5.2 Ingition delay. Ignition delay is the time measured between the firing pin striking the primer or ignition device and the first indication of pressure.

6.6 Safety precautions. The safety precaution requirements of "Contractor' Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract as required by the Federal Acquisition Regulation (FAR) 23.3. NOTE: When this specification is used as part of the description of work to he accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP 5) should be made applicable.

6.7 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

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6.8 **Changes from previous issue.** Asterisks are not used in this revision to identify changes with respect to the previous issue because of the extensiveness of the changes.

Custodian:
Navy-OS
Air Force-99

Preparing activity:
Navy-OS
(Project 1377-0958)

Review activity:
Air Force-70

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INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-I-46625C	900131

3. DOCUMENT TITLE

INITIATOR, CARTRIDGE ACTUATED, M3A2, ASSEMBLY

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Commanding Officer Naval Ordnance Station	(1) Commercial (301) 743-4250	(2) AUTOVON 364-4250
c. ADDRESS (Include Zip Code) (Code 3730) Indian Head, MD 20640-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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